

Relational Algebra Practice Problem

Given Schema:

Spring 2025

1. Hero (**hero_id(pk)**, hero_name, real_name, species, homeworld)
2. Team (**team_id(pk)**, team_name, **leader_id(fk)**, headquarters)
3. Mission (**mission_id(pk)**, mission_name, location, date, success)
4. Assignment (**hero_id(fk)(pk)**, **mission_id(fk)(pk)**, role)
5. Find the relational Algebra for the followings:

- a) Find the names of all heroes who are from Earth.

MySQL:

```
SELECT hero_name
FROM Hero
WHERE homeworld= 'Earth';
```

- b) List the names of all Avengers who have participated in at least one mission.

MySQL:

```
SELECT hero_name
FROM Hero H
JOIN Assignment A
ON H.hero_id = A.hero_id
```

Or

```
SELECT DISTINCT H.hero_name
FROM Hero H
JOIN Team T ON H.hero_id = T.leader_id
JOIN Assignment A ON H.hero_id = A.hero_id
WHERE T.team_name = 'Avengers';
```

- c) Find the names of team leaders whose teams have their headquarters in Wakanda.

MySQL:

```
SELECT H.hero_name
FROM Hero H JOIN Team T ON H.hero_id = T.leader_id
WHERE T.headquarters = 'Wakanda';
```

Fall 2024

1. Patients (**patient_id**, name, age, gender, address)
2. Doctors (**doctor_id**, name, specialization, salary)
3. Appointments (**appointment_id**, **patient_id**, **doctor_id**, date, time_slot)
4. Medicine (**medicine_id**, name, category, price)
5. Medications (**prescription_id**, **medicine_id**, dosage)
6. Prescriptions (**prescription_id**, **appointment_id**, remarks)

Answer the following questions using relational algebra:

- a) Find the details of male patients who reside in Palashi.

MySQL:

```
SELECT *
FROM Patients
WHERE gender = 'male' AND address = 'Palashi';
```

b) List the names of all patients along with the medicines prescribed to them.

MySQL:

```
SELECT P.name, M1.name
FROM Patients P
JOIN Appoints A ON P.patient_id = A.patient_id
JOIN Prescriptions P1 ON A.appointment_id = P1.appointment_id
JOIN Medications M ON P1.prescription_id = M.prescription_id
JOIN Medicine M1 ON M1.medicine_id = M.medicine_id;
```

c) Find the specializations of doctors who have never treated patients older than 70 years.

MySQL:

```
SELECT DISTINCT D.specialization
FROM Doctors D
LEFT JOIN Appointments A ON D.doctor_id = A.doctor_id
LEFT JOIN Patients P ON A.patient_id = P.patient_id AND P.age > 70
WHERE P.patient_id IS NULL;
```

Or with subquery:

```
SELECT DISTINCT D.specialization
FROM Doctors D
WHERE D.doctor_id NOT IN (
    SELECT DISTINCT A.doctor_id
    FROM Appointments A
    JOIN Patients P ON A.patient_id = P.patient_id
    WHERE P.age > 70
);
```